

# A four-vertex counterexample to Knudson's persistence-gradient conjecture

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## Abstract

Knudson conjectured that every nonincident persistence pair is joined by exactly one gradient path in the discrete vector field obtained by retaining the incident persistence pairs. We give a simplex-wise filtration of a triangulated interval with four vertices and three edges for which one such pair has no gradient path. This disproves Conjecture 2 in his contribution to Oberwolfach Report 29/2008. The proof consists of three boundary reductions and an inspection of the matching. The same example works over any field.

## 1 The statement and the example

Let  $P$  be the persistence pairing of a finite simplicial complex whose filtration adds one simplex at a time, with coefficients in  $\mathbb{F}_2$ . For a codimension-one face relation  $\sigma < \tau$ , put

$$V_P = \{(\sigma, \tau) \in P : \sigma < \tau\}.$$

Knudson's Conjecture 2 [1, p. 1629] predicts a unique gradient path in this fixed field for every nonincident pair in  $P$ . We use the directed incidence graph: unmatched incidences point from a simplex to a facet, and matched incidences point upward. A gradient path from a critical edge starts at an endpoint and then follows matched upward steps alternating with downward steps to the other endpoint.

Take vertices  $a, b, c, d$  and edges  $cd, bd, ac$ , with no higher simplices. Here  $xy$  denotes  $\{x, y\}$ . The underlying space is the interval  $a-c-d-b$ . Insert the seven nonempty simplices in the order

$$a, b, c, d, cd, bd, ac. \tag{1}$$

Every face appears before its coface, so this is an admissible filtration. The indices  $1, \dots, 7$  are filtration values, not a discrete Morse function whose own gradient is claimed to equal  $V_P$ .

**Theorem 1.** *For the filtration (1), the finite persistence pairs and their incident submatching are*

$$P = \{(d, cd), (c, bd), (b, ac)\}, \quad V_P = \{(d, cd)\}.$$

*The pair  $(b, ac)$  is nonincident, and there are zero gradient paths from  $ac$  to  $b$  in  $V_P$ . Consequently Knudson's Conjecture 2 is false.*

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*Proof.* Order the vertex rows as  $a < b < c < d$  and edge columns as  $cd < bd < ac$ . Over  $\mathbb{F}_2$ , the usual left-to-right persistence reduction [2] eliminates repeated latest nonzero rows using earlier reduced columns:

$$\begin{aligned} R(cd) &= \partial(cd) = c + d, \\ R(bd) &= \partial(bd) + R(cd) = (b + d) + (c + d) = b + c, \\ R(ac) &= \partial(ac) + R(bd) = (a + c) + (b + c) = a + b. \end{aligned}$$

The three pivots are respectively  $d, c, b$ , giving precisely the stated pairs. The vertex  $a$  is unpaired. Of those pairs, only  $d < cd$  is an incidence, giving the stated  $V_P$ .

For an independent check of the pairing,  $cd$  kills the component born at  $d$ ;  $bd$  merges the component born at  $c$  into the older one born at  $b$ ; and  $ac$  merges that surviving component into the oldest one, born at  $a$ . This is the same pairing in ordinary  $H_0$ .

The complete directed incidence graph has the arrows

$$ac \rightarrow a, \quad ac \rightarrow c, \quad bd \rightarrow b, \quad bd \rightarrow d, \quad d \rightarrow cd, \quad cd \rightarrow c.$$

It is acyclic. In particular  $V_P$  is a discrete gradient, and both  $b$  and  $ac$  are critical. Either first step from  $ac$  reaches  $a$  or  $c$ ; both are unmatched vertices, so the path stops. The reachable set is exactly  $\{ac, a, c\}$ , which excludes  $b$ .  $\square$

## 2 Why cancellation does not repair the fixed field

The reduction of  $ac$  uses  $R(bd)$  to remove its pivot at  $c$ . But the persistence pair  $(c, bd)$  is nonincident and contributes no upward step to  $V_P$ . Algebraic column elimination therefore need not trace a path in that field.

There is a unique path for the earlier pair  $(c, bd)$ :

$$bd \rightarrow d \rightarrow cd \rightarrow c.$$

Reversing it gives the new matching  $V' = \{(d, bd), (c, cd)\}$ , in which the path

$$ac \rightarrow c \rightarrow cd \rightarrow d \rightarrow bd \rightarrow b$$

now exists. This observation concerns  $V'$ , whereas the conjecture specifies  $V_P$ . It is consistent with the role of ordered cancellations in the surface simplification results of Bauer, Lange, and Wardetzky [3, Lemma 9]; we claim no counterexample to those results.

**Other coefficients.** Orient each edge from its alphabetically earlier to its later vertex. Over any field the identical elimination, with subtraction, gives

$$R(cd) = d - c, \quad R(bd) = (d - b) - (d - c) = c - b, \quad R(ac) = (c - a) - (c - b) = b - a.$$

All three pivot coefficients are 1. The pairs, matching, and absence of a path are unchanged. No minimality claim is needed or made.

**Reproducibility and status.** The companion package provides two standard-library Python verifiers: one uses boundary reduction and union-find; the other reconstructs the  $H_0$  rank invariant from component partitions and also checks signed reduction and path reversal. The proof above is independent of these programs. A documented literature audit completed on 23 September 2026 located no earlier refutation of the stated conjecture. This is a bounded search, not a guarantee of priority; in particular, an author bibliography lists related work in preparation. The audit records inaccessible sources. This unrefereed note and its checks were prepared with OpenAI Codex and separate AI audit agents, from a candidate construction supplied by the author from prior AI-assisted work.

## References

- [1] K. P. Knudson, Persistent homology and discrete Morse theory, in *Computational Algebraic Topology*, Oberwolfach Report 29/2008, *Oberwolfach Reports* **5** (2008), 1628–1630. Conjecture 2, p. 1629. [doi:10.4171/OWR/2008/29](https://doi.org/10.4171/OWR/2008/29).
- [2] H. Edelsbrunner, D. Letscher, and A. Zomorodian, Topological persistence and simplification, *Discrete Comput. Geom.* **28** (2002), 511–533. [doi:10.1007/s00454-002-2885-2](https://doi.org/10.1007/s00454-002-2885-2).
- [3] U. Bauer, C. Lange, and M. Wardetzky, Optimal topological simplification of discrete functions on surfaces, *Discrete Comput. Geom.* **47** (2012), 347–377. [doi:10.1007/s00454-011-9350-z](https://doi.org/10.1007/s00454-011-9350-z).